

$$\text{equation}(A) \quad \theta'' = -\frac{3g}{2l} \cos(\theta)$$

$$\text{equation}(B) \quad \begin{cases} x(t) \equiv x_c + t - 5 \cos(\theta) \\ y(t) \equiv 5 \sin(\theta) \end{cases}$$

The ground is assumed to be parallel to the tangent of the surface of the Earth
 $\theta \equiv$ the angle that the ladder forms with the ground

$\omega \equiv$ the angular velocity of the ladder

$\alpha \equiv$ the angular acceleration of the ladder

And the angle between the ladder and the floor changes with the time in accordance with the equation

$$\theta(t) = \arccos\left(\frac{X(t)}{L}\right) \quad (1.1)$$

where $X(t)$ is given by $X(t) = x_0 + t \cdot V(t)$ and $V(t)$ is given as 1 m/s Because $\omega = \lim_{x \rightarrow a} \frac{\Delta\omega}{\Delta t} = \frac{d\theta}{dt}$, the angular velocity of the falling end of the ladder can be determined by

$$\omega(t) = -\frac{V(t) + t \frac{dV}{dt}}{L \sqrt{1 - \frac{(x_0 + tV(t))^2}{L^2}}} \quad (1.2)$$

Furthermore, since $\alpha \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt}$, the angular acceleration of the falling end of the ladder can be given by

$$\alpha(t) = -\frac{(x_0 + tV(t)) \left(V(t) + t \frac{dV}{dt}\right)^2 + L^2 \left(1 - \frac{(x_0 + tV(t))^2}{L^2}\right) \left(2 \frac{dV}{dt} + t \frac{d^2V}{dt^2}\right)}{L^3 \left(1 - \frac{(x_0 + tV(t))^2}{L^2}\right)^{\frac{3}{2}}} \quad (1.3)$$

A little more manipulation gives us

$$V_x(t) = \left(\frac{k}{m}t + \frac{1}{v_{0x}}\right)^{-1} \quad (16)$$

This gives us the velocity of the ball in the x-direction at time t. We also want the position at time t, which we derive by integrating:

$$p_x(t) = \int v_x dt = \int \left(\frac{k}{m}t + \frac{1}{v_{0x}}\right)^{-1} dt = \frac{m}{k} \ln|kv_{0x}t + m| \quad (17)$$

In the y-direction, equations must take into account both air friction and gravity. We can start the same way as with the x-direction:

$$F_y = g - kv_y^2 = ma_y = m \frac{dv_y}{dt} \quad (18)$$